Emanuel Camacho

MSc Candidate in Aerospace Engineering • Aerodynamics, Propulsion & Control Systems • Available from January 2026
emanuel.aerospace@gmail.com | +351 960 366 633
linkedin.com/in/emanuel-camacho-05ab8a245 | emanuelcamaerospace.github.io/website

EDUCATION

Instituto Superior Técnico — Expected Nov 2026
MSc in Aerospace Engineering — *Lisbon, Portugal*
- Specialisation in Aerodynamics and Aircraft Propulsion Systems.

Technical University of Munich (TUM) — Aug 2025
Erasmus Exchange Semester, MSc in Aerospace Engineering — *Munich, Germany*
- Coursework in aircraft propulsion and aircraft system integration.

Instituto Superior Técnico — Jul 2024
BSc in Aerospace Engineering — *Lisbon, Portugal*
- Relevant coursework: Fluid Dynamics, Flight Mechanics, Flight Control.

RESEARCH

Instituto Superior Técnico & Portuguese Air Force — Sep 2025 – Nov 2026
MSc Thesis: Low-Power Laser Neutralization of UAV Optical Sensors — *Lisbon, Portugal*
- **Research question:** Assessed whether a low-power laser on a low-cost 2-axis gimbal can be held on a UAV's camera precisely and long enough to dazzle or damage it. Funded by the Portuguese Air Force, in collaboration with the Institute for Systems and Robotics (ISR).
- **Control:** Derived the coupled gimbal dynamics (Newton–Euler and Lagrange) and designed an Integral Sliding Mode Controller with Lyapunov stability analysis, tuning its gains with a genetic algorithm in simulation and directly on the hardware.
- **Simulation:** Built an end-to-end MATLAB/Simulink and Simscape Multibody model (generated from CAD) with YOLO-based detection and EKF state estimation, IMU noise, servo dynamics and a Gaussian-beam laser energy model; identified gimbal parameters by grey-box system identification.
- **Prototype:** After a two-month delay and partial hardware delivery, built a working prototype from 3D-printed parts, a Pixhawk and hobby-grade servos in about four weeks, with real-time control from Simulink via Python/MAVLink.
- **Results:** Achieved median tracking errors of 0.56° (yaw) and 0.29° (pitch) on a UAV at about 3 m, limited mainly by the 0.45° servo dead band; showed that at short range, pointing precision rather than laser power is the limit.

EXPERIENCE

AeroTéc – BLUE Turbojet Project — Jan 2026 – Present
Design Team Member, Compressor Section — *Lisbon, Portugal*
- Carry out the aerodynamic design and optimisation of the centrifugal impeller (90,000 rpm) and diffuser of a small-scale turbojet engine, targeting a compressor-stage pressure ratio of 3.5, within a 15-member team.
- Developed MATLAB thermodynamic and mean-line flow scripts to compute 3D blade geometry, flow paths and stage pressure ratios.
- Built the team's automated parametric CAD framework by linking MATLAB to Siemens NX, reducing each design iteration to a single click and around 30 seconds of processing.
- Perform preliminary 3D CFD simulations in Siemens STAR-CCM+ to assess boundary-layer separation, optimise blade incidence angles and verify predicted pressure ratios.

ANAC – Portuguese National Civil Aviation Authority — Aug 2024 – Sep 2024
Project Intern — *Lisbon, Portugal*
- Developed models predicting the post-failure descent of commercial UAS, computing ballistic trajectories and ground-impact zones after critical system failures.
- Built an interactive browser-based simulation tool that allows certification engineers to assess UAS safety parameters.

SKILLS & LANGUAGES

Software: Siemens NX, SolidWorks, Solid Edge (CAD); Siemens STAR-CCM+ (CFD); MATLAB, Simulink, Simscape; Python, C; LaTeX.
Engineering: Turbomachinery Design, Aerodynamic Analysis, Computational Fluid Dynamics, Nonlinear Control (Sliding Mode), System Identification, C-UAS System Architecture, Systems Engineering.
Languages: Portuguese (Native), English (C2), Spanish (C1), German (A2).

CERTIFICATIONS

STAR-CCM+ CFD Training Certification – Cadflow / Siemens Digital Industries Software — Sep 2026
German Language, Level A2 – Volkshochschule Lernportal (A1.1 at TUM, Aug 2025) — Aug 2026